2501-PYL1003 (Mechanics & STR) — Practice Mid-Sem Paper 4

Full Marks: 50 | Time: 2 hours | Instructor Style: Pradipta Ghosh (IIT Delhi)

Instructions

An MCQ can also be an MSQ (multiple correct options). No credit without justification/derivation.
Show all intermediate steps. Numerical answers without units lose marks.
Approximations must be justified explicitly.

Q1. A student claims that the period τ of small oscillations of a simple pendulum depends on its length ℓ, the acceleration due to gravity g, and the mass m of the bob, i.e., $\tau = C\,\ell^a g^b m^c$ for some dimensionless constant C.

(i) Show, using dimensional analysis, that the exponent c **must** be zero — i.e., that τ cannot depend on m — and find a and b. **(3)**

(ii) A different student argues this proves mass has "no effect on any pendulum's motion whatsoever." Give one physical scenario (still a pendulum-like system) where the period **does** depend on a mass-like quantity, and briefly explain why the dimensional argument in (i) does not apply there. **(2)**

Q2 (MSQ). In the CGS (centimeter-gram-second) system, the unit of force is the dyne ($1 \text{ dyne} = 10^{-5} \text{ N}$) and the unit of energy is the erg ($1 \text{ erg} = 10^{-7} \text{ J}$). Which of the following is/are correct? **(1+1+1=3)**

a. $1 \text{ erg} = 1 \text{ dyne} \times 1 \text{ cm}$. b. A quantity with SI value 6 J has the numerical value 6×10^7 in erg. c. Pressure measured as 2 Pa equals 20 dyne/cm^2. d. The dimensional formula of a physical quantity changes when switching from SI to CGS units.

Q3. A particle moves along the x-axis with jerk (rate of change of acceleration) held constant: $\dddot{x} = j_0$ (a constant). At $t = 0$: $x = 0, v = 0, a = -a_0$ (with $a_0 > 0$).

(i) Find $a(t), v(t)$, and $x(t)$ explicitly. **(3)**

(ii) Find the time t^* at which the acceleration becomes zero, and determine whether the velocity is at a maximum, minimum, or neither at that instant. Justify using the sign of $\dot{a}$ around t^*. **(3)**

Q4. A boat of mass M (including a passenger) moves through still water. The water exerts a resistive force on the boat proportional to its instantaneous speed, $F_{\text{resist}} = -bv$. Starting at $t = 0$ with speed v_0, the engine is turned off (no driving force thereafter).

(i) Find $v(t)$. **(2)**

(ii) Find the total distance D the boat travels before (theoretically) coming to rest, by integrating $v(t)$ from $t = 0$ to ∞. **(2)**

(iii) A second, identical boat starts with the same v_0 but experiences resistance $F = -b'v^2$ instead (quadratic drag). Without fully solving for $x(t)$, determine whether this boat travels a **finite** or **infinite** total distance before stopping, and justify with a brief argument (e.g., examine $v(x)$ via $a = v\,dv/dx$). **(3)**

Q5. A small block of mass m is placed at the top of a frictionless hemispherical dome of radius R fixed to the ground, and given a negligible push so it starts sliding down the outside surface.

(i) Using energy conservation, find the speed v of the block as a function of the angle θ measured from the top of the dome (i.e., from the vertical through the center). **(2)**

(ii) Find the angle θ_c at which the block leaves the surface of the dome (i.e., where the normal force becomes zero). **(4)**

Q6. A point P is specified in spherical polar coordinates as $r = 20, \theta = 90^\circ, \phi = 210^\circ$.

(i) Find the Cartesian coordinates of P exactly (in terms of surds if necessary, or to two decimal places). **(2)**

(ii) A second point Q has cylindrical coordinates $\rho = 20, \phi = 210^\circ, z = 0$. Show, with reasoning (not just by comparing numbers), why P and Q **must** coincide given the value of θ at P. **(2)**

(iii) Find the straight-line (Cartesian) distance between P and a third point S with cylindrical coordinates $\rho = 20, \phi = 30^\circ, z = 0$. **(2)**

Q7. Consider a particle constrained to move on a cone of half-angle α (i.e., $\theta = \alpha$ is fixed, but $r(t)$ and $\phi(t)$ both vary) under some unspecified force.

(i) Using $\vec{v} = \dot{r}\,\hat{r} + r\dot{\theta}\,\hat{\theta} + r\sin\theta\,\dot{\phi}\,\hat{\phi}$ with $\theta = \alpha$ constant, write down $\vec{v}$ and the kinetic energy $T = \frac{1}{2}m|\vec{v}|^2$ in terms of $r, \dot{r}, \dot{\phi}, \alpha$. **(2)**

(ii) Using $\vec{a} = (\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\,\dot{\phi}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\sin\theta\cos\theta\,\dot{\phi}^2)\hat{\theta} + \ \frac{1}{r}\frac{d}{dt}(r^2\sin\theta\,\dot{\phi})\ \ \hat{\phi}$ specialized to $\theta = \alpha$ constant, show that the $\hat{\theta}$-component of acceleration reduces to $-r\sin\alpha\cos\alpha\,\dot{\phi}^2$, and give a brief physical interpretation of this term (it must be balanced by the normal force from the cone's surface). **(4)**

Q8. A vector $\vec{A} = 3\hat{\imath} - 4\hat{\jmath} + 12\hat{k}$ is defined at the point $(x, y, z) = (3, -4, 12)$.

(i) Express $\vec{A}$ in terms of the spherical unit vectors $\hat{r}, \hat{\theta}, \hat{\phi}$ **at that point** (i.e., find the components A_r, A_θ, A_ϕ). Round to two decimal places where needed. **(5)**

(ii) Verify your answer by confirming that $|\vec{A}|$ computed from the spherical components matches $|\vec{A}|$ computed from the Cartesian components. **(1)**

Q9. A particle of mass m moves under a central force $F(r) = -\frac{k}{r^2} + \frac{c}{r^3}$ (an inverse-square force perturbed by a small inverse-cube term, c small and positive).

(i) Show that the effective potential $U_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{k}{r} + \frac{c'}{r^2}$ (for a suitable constant c' related to c) can be written in the same functional form as a pure inverse-square problem but with a **modified** effective angular momentum L_{eff}. Find L_{eff} in terms of L, m, c. **(4)**

(ii) Using this substitution, argue (without re-deriving the full orbit equation from scratch) that the orbit is still a conic section in the variable $u = 1/r$, but that the apsidal angle (angle swept between successive perihelion passages) is **not** 2π in general. State whether the apsidal angle is greater than or less than 2π for $c > 0$, with brief reasoning. **(3)**

Q10. Kepler's second law states that a planet sweeps out equal areas in equal times.

(i) **Prove** Kepler's second law from Newton's second law for **any** central force (not just inverse-square), starting from $\vec{F} = F(r)\hat{r}$ and showing that $\vec{L} = m\vec{r} \times \vec{v}$ is conserved. **(3)**

(ii) A planet's orbital speed at a point where its radius vector makes angle 90° with its velocity is v_1, at radius r_1. At another point in the orbit, its speed is v_2 at radius r_2, with the radius vector making angle β (not 90°) with the velocity there. Write down the correct relation between $r_1, v_1, r_2, v_2, \beta$ using conservation of angular momentum (be careful with the geometric factor). **(2)**

Q11. A system consists of a fixed pulley at the top of a frictionless incline of angle θ. A block of mass m_1 rests on the incline, connected via a string over the pulley to a block of mass m_2 hanging vertically off the other side (off the top of the incline, in free space). Additionally, a horizontal force F is applied to m_1, directed **into** the incline (i.e., horizontally, pushing m_1 up the slope direction's horizontal component).

(i) Draw the FBD of m_1 or m_2, resolving F into components along and perpendicular to the incline. **(2)**

(ii) Derive the condition on F (in terms of m_1, m_2, θ, g) required to hold the system in static equilibrium (assume the incline is frictionless, so equilibrium requires exact force balance along the string direction). **(5)**

Q12. A particle of mass m moves in a 1D potential $U(x) = U_0 \left(\frac{x^2}{a^2} - 1 \right)^2$ (a symmetric double-well potential), where $U_0, a > 0$ are constants.

(i) Find all equilibrium points and classify each as stable/unstable. **(3)**

(ii) The particle is released from rest at $x = 2a$. Find the speed of the particle as it crosses $x = 0$, in terms of U_0, a, m. **(3)**

Q13. A thin uniform rod of mass M and length L lies on a frictionless horizontal table. A point particle of mass m, moving with speed v_0 perpendicular to the rod, strikes and sticks to one end of the rod (perfectly inelastic collision).

(i) Find the velocity of the center of mass of the combined system immediately after the collision. **(2)**

(ii) Find the angular velocity of the combined (rod + particle) system about its **new** center of mass immediately after the collision. [The moment of inertia of a uniform rod about its center is $ML^2/12$; use the parallel axis theorem as needed.] **(4)**

Q14. For a system of N particles with masses m_i, position vectors $\vec{r}_i$ (from a fixed origin O), and velocities $\vec{v}_i$, define the center of mass $\vec{R} = \dfrac{1}{M}\sum_i m_i \vec{r}_i$ where $M = \sum_i m_i$.

Prove the following decomposition of total kinetic energy (König's theorem):

$$T = \frac{1}{2} M \dot{\vec{R}}^2 + \frac{1}{2} \sum_i m_i \dot{\vec{r}}_i'^2$$

where $\vec{r}_i' = \vec{r}_i - \vec{R}$ is the position of particle i relative to the center of mass. Show every cross-term explicitly and explain why one of them vanishes identically. **(6)**

Q15. A block of mass $6\ \mathrm{kg}$ is held against a vertical wall by a horizontal force $F = 80\ \mathrm{N}$ pressing it into the wall. The coefficient of static friction between block and wall is $\mu_s = 0.5$.

(i) Determine whether the block remains stationary (does not slide down) under gravity alone, given $g = 10\ \mathrm{m/s}^2$. Show the relevant inequality clearly. **(2)**

(ii) If an additional constant vertical force of $10\ \mathrm{N}$ **downward** is now also applied to the block (in addition to F and gravity), re-determine whether the block still remains stationary. **(2)**

(End of Paper 4 — 50 marks)